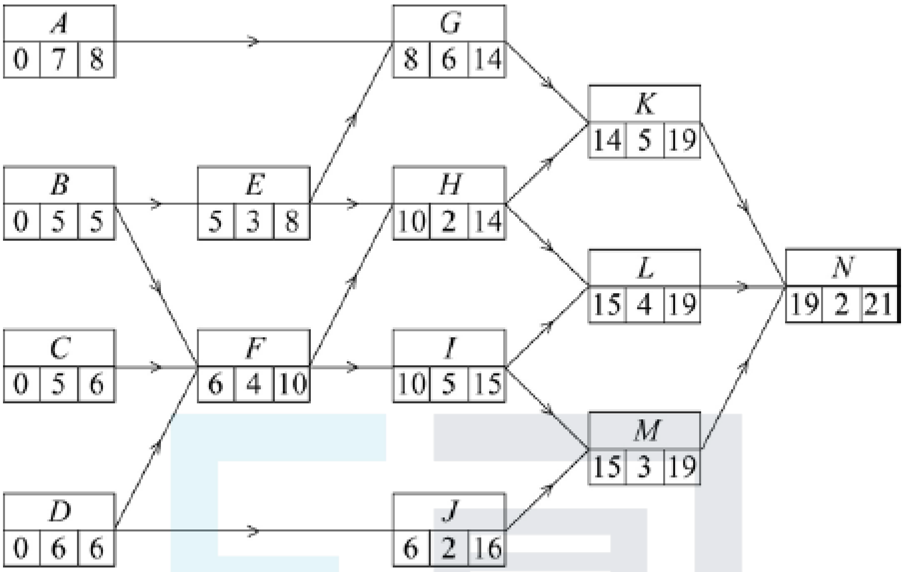
Mark schemes

Q1.

(a)



Forward pass

condone one slip (follow through)

M1

all correct

A1

Backward pass

condone one slip (follow through)

M1

all correct

A1

4

(b) Critical paths

$B\ E\ G\ K\ N$

first path correct

M1

$D\ F\ I\ L\ N$

second path and no others

A1

Minimum completion time is 21 days

B1

3

(c) Cascade diagram

may be in blocks or bars (see examples)

One of 'their' CPs correct

ft their CP

M1

$B , D , E , F , G , I , K , L , N$

these activities correct

A1

A , C , H , J , M

3 of these with correct start and duration

M1

3 correct with correct slack indicated

A1

all 5 correct with correct slack

A1

		Slack
A	0 − 7	7 − 8
C	0 − 5	5 − 6
H	10 − 12	12 − 14
J	6 − 8	8 − 16
M	15 − 18	18 − 19

5

(d) (Max value of x is) 10

considering $J_{latest} - J_{earliest}$

M1

$\Rightarrow x \leq 10$

(condone $x < 11$ for SC2)
NMS $x \leq 10$ award M1 A1

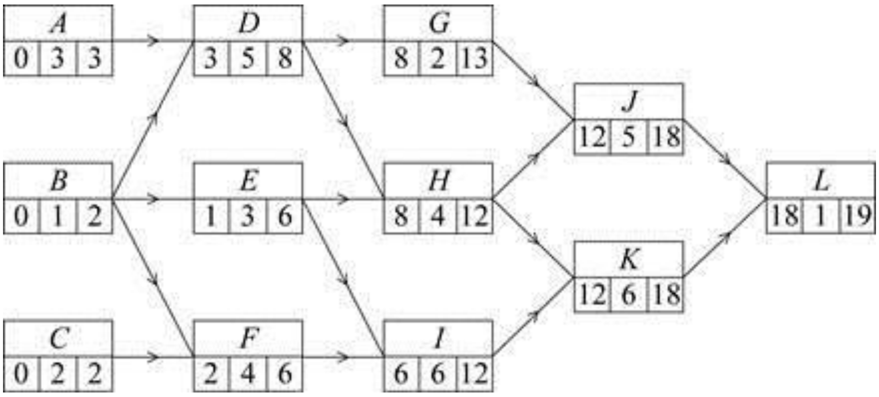
A1 cao

2

[14]

Q2.

(a)



Earliest start times

condone one slip + ft

M1

all correct

A1

Latest finish times

condone one slip + ft

M1

all correct

A1

4

(b) Critical paths $A\,D\,H\,K\,L$

one path correct

B1

$C\,F\,I\,K\,L$

second path correct and no others

B1

Minimum time = 19

19 days

B1

3

(c) Greatest float time at G

ft their activity with greatest float for M1

M1

(13 – 8 – 2) = 3 (days)

values at G must be correct

A1cso

2

(d) *one of 'their' critical paths "correct"*

M1

A, D, H, K, L and C, F, I

all 8 of these activities correct

A1

B, E, G, J

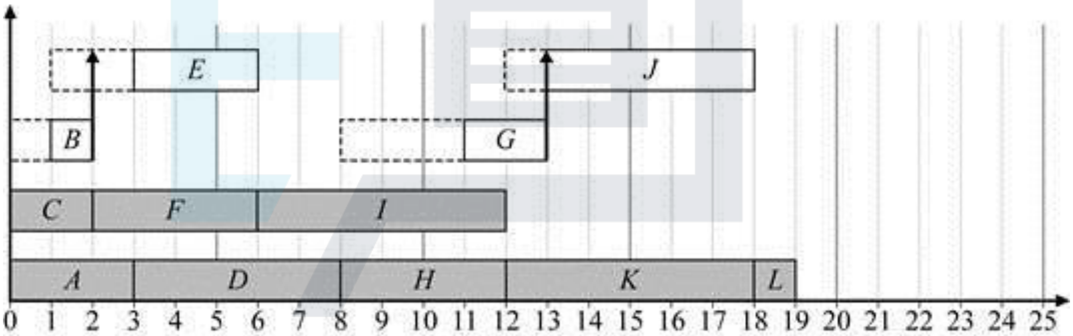
B(1-2); E(3-6); G(11-13); J(13-18)

3 of these with correct duration and latest start time (may omit slack)

M1

all 4 correct with correct slack shown

A1cso

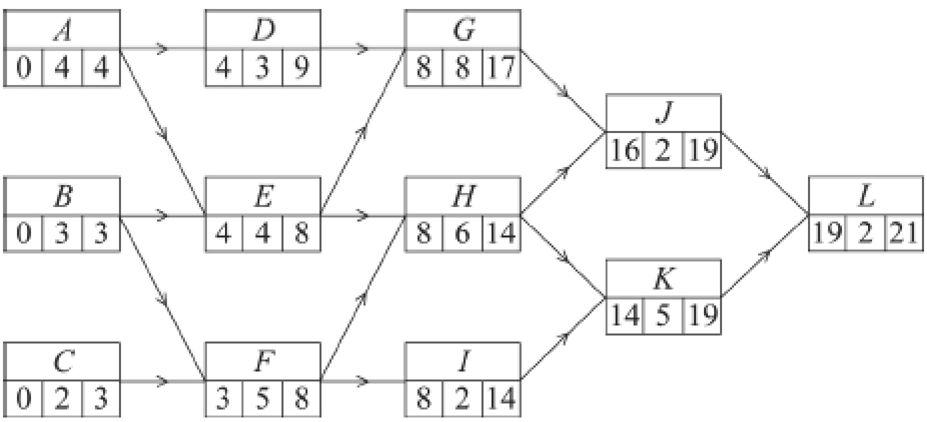


4

[13]

Q3.

(a)



Earliest start times

one slip follow through

M1

all correct

A1

Latest finish times

one slip follow through

M1

all correct

A1

4

(b) Critical paths are $AEHKL$ and $BFHKL$

one correct

M1

both correct and no extras

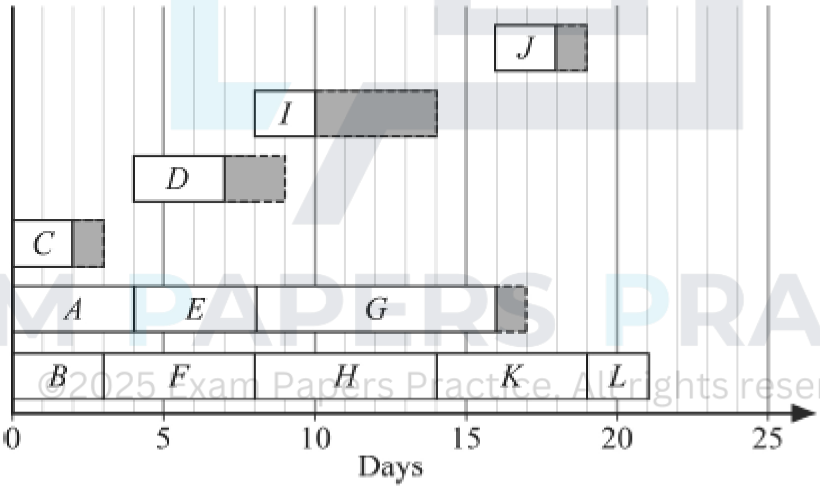
A1

Minimum completion time = 21 days

B1

3

(c)



A (0 → 4)
B (0 → 3)
C (0 → 2 → 3)
D (4 → 7 → 9)
E (4 → 8)
F (3 → 8)
G (8 → 16 → 17)
H (8 → 14)
I (8 → 10 → 14)
J (16 → 18 → 19)
K (14 → 19)
L (19 → 21)

A, B, E, F, H, K, L correct

B1

C, D, G, I, J
(4 with correct start and duration)

M1

All 5 correct with correct slack indicated

A1

3

(d) (i) *K* now starts day 17
or "delayed" b 3 days if 14 in network

B1

L now starts day 22
or "delayed" b 3 days if 19 in network

B1

2

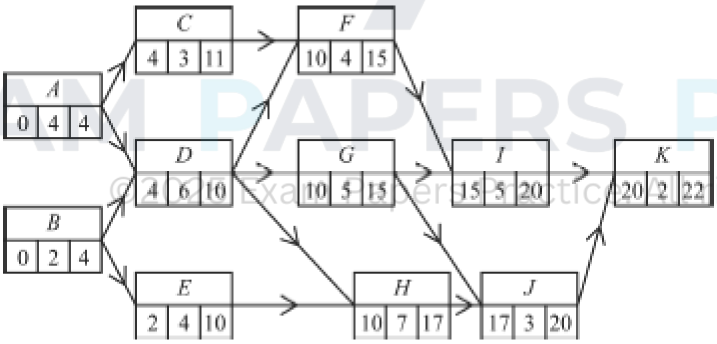
(ii) Overall delay 3 days

B1

1

[13]

Q4.



(a) Earliest start times
up to 2 errors ft

M1

All correct

A1

Latest finish times
up to 2 errors ft from K

M1

Correct

A1

4

(b) Critical paths $A\ D\ G\ I\ K$

$A\ D\ H\ J\ K$

withhold if extra path such as
$A\ D\ G\ J\ K$ given

B1B1

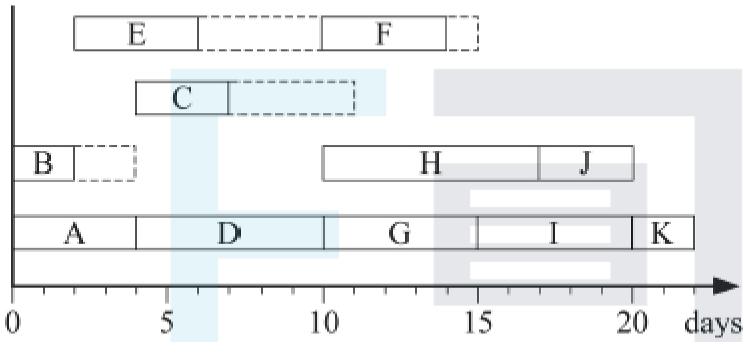
Minimum time for completion = 22 days

Must be stated – not just a value in box

B1

3

(c)



A, D, G, H, I, J, K correct

B1

B, C, E, F

2 correct with slack /float or 4 correct & no slack

B1

all correct with slack/float; withhold if
slack not shown dotted etc

B1

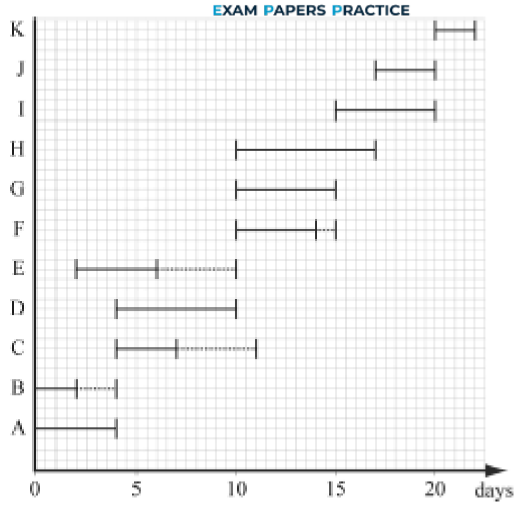
3

(d) F starts after 12 days at earliest
or F starts 2 days later
I is now unable to start until after 16 days
or starts 1 day later

E1

Minimum completion time now 23 days – one extra day etc

B1

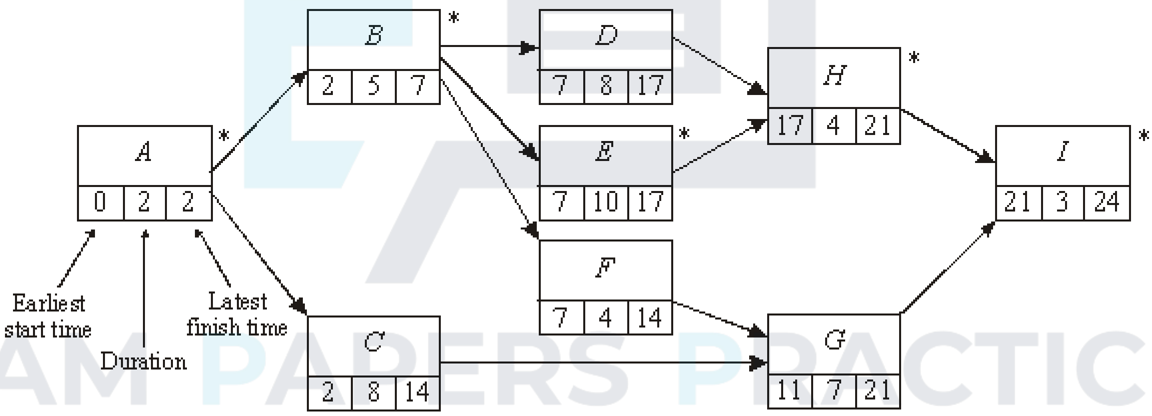


2

[12]

Q5.

(a)



SCA

M1

(almost correct 2 slips)

A1

Correct

A1

3

(b) Forward pass for earliest start times

M1

All correct

A1

2

(c) Backward pass for latest finish times

M1

All correct

A1

2

(d) Critical path $A\ B\ E\ H\ I$

B1

1

(e)

Non critical	C	D	F	G
Float	4	2	3	3

At least one float time correct

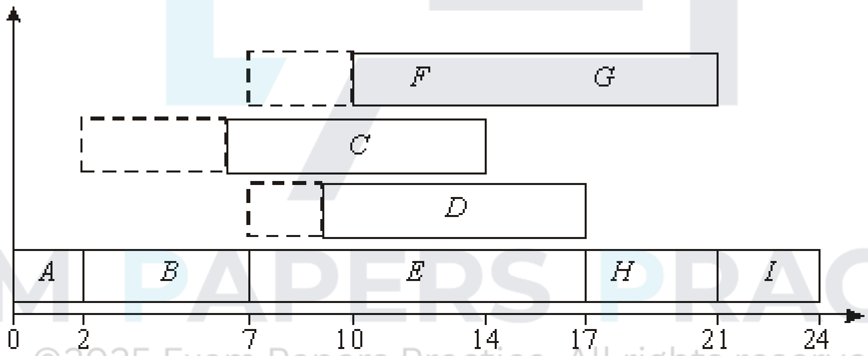
M1

All correct

A1

2

(f)



'their' critical path on chart

B1ft

C from 6 to 14 (with space 2-6)

One other activity (condone no slack or earliest start)

M1

D from 9 to 17 (with slack 7-9)

2 other non critical activities

A1

F & G from 10 to 21 with appropriate slack

All correct

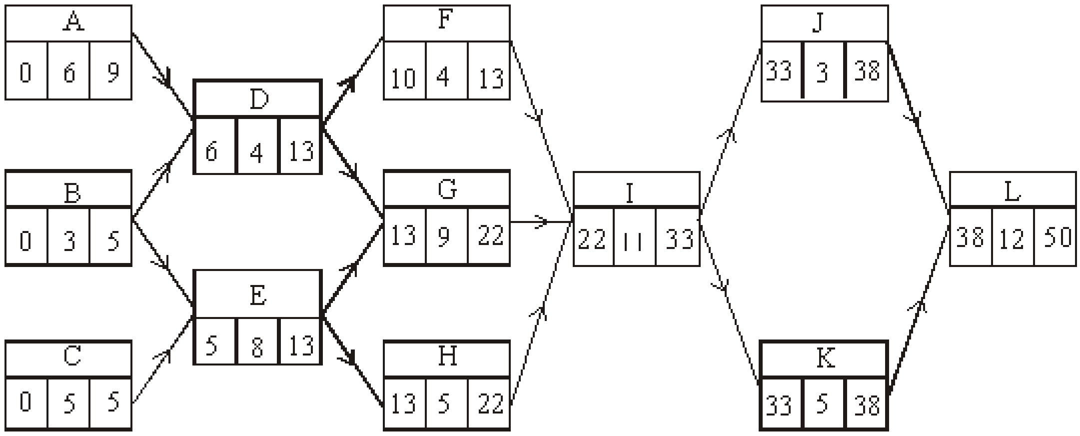
A1

4

[14]

Q6.

(a)



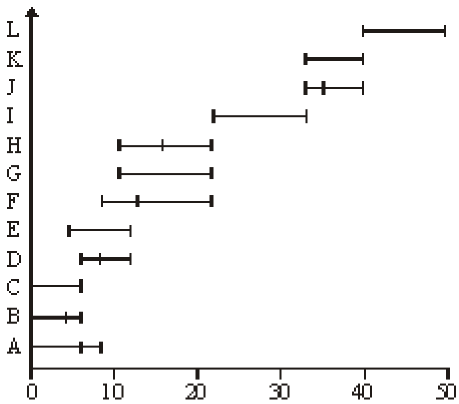
forward pass M1A1 **2**

back pass M1A1 **2**

(b) *C E G I K L* B1 **1**

(c) *F* B1 **1**

(d)



Gantt diagram M1

floats included

correct, excluding floats B1 A1 **3**

(e) (i) D 5 days $\Rightarrow$ G back 2 days E1

∴ J starts at 35
OE M1

∴ L starts at 43

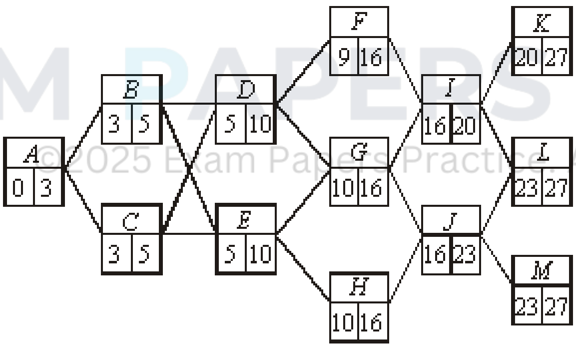
∴ finish at 55 A1 **3**

(ii) $A\ D\ G\ I\ J\ L$ B1 **1**

[13]

Q7.

(a) (i) & (ii)



Forward M1 A1 **2**

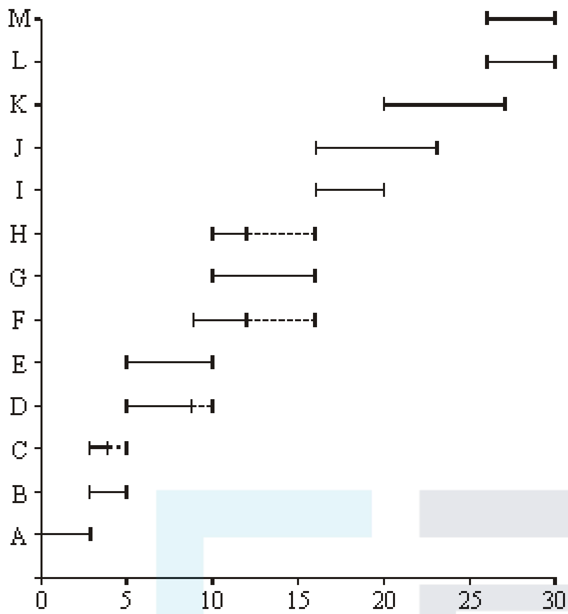
Back M1 A1 **2**

(b) C, H, F, D

B1

1

(c)



M1

(–1 EE)

A2

3

(d) (i) Extra at KLM

M1

Min extra 4
Total = 31

A1

2

(ii) A B C D E F G H J I L K M

B1

OE

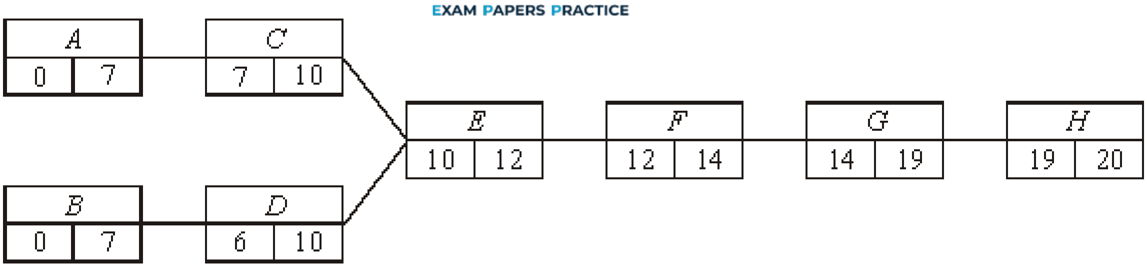
B1

2

[12]

Q8.

(a)



–1 EE

M1 A2

3

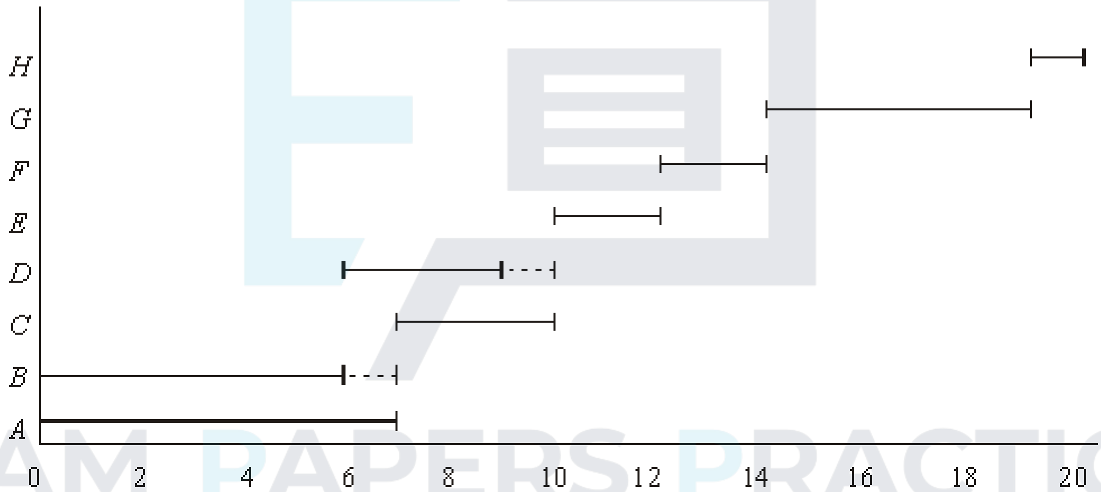
(b) Min = 20

M1 earned in (a)

M1 A1

2

(c)



–1 EE

M1 A3F

4

[9]

Q9.

(a)

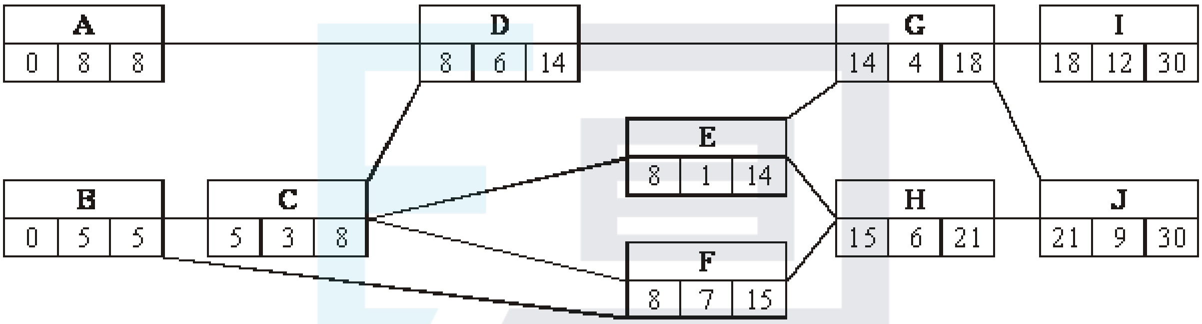
Activity	Immediate Predecessor
A	-
B	-
	B
C	A, C
D	C

E	(B,) C
F	D, E
G	E, F
H	G
I	G, H
J	

–1 EE

B2, 1 **2**

(b)



Forward pass

M1 A1

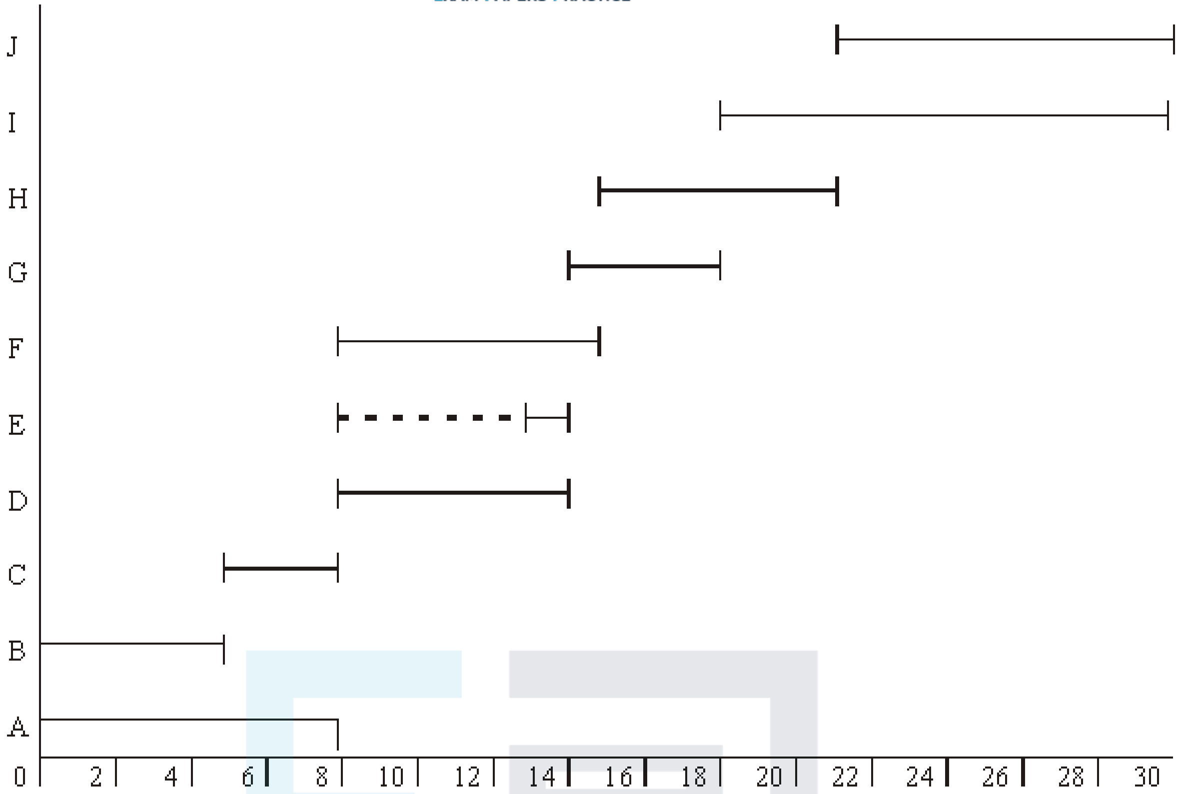
Back pass

M1 A1 **4**

(c) Critical A, B, C, D, F, G, H, I, J

B1 **1**

(d)



−1 EE

M1 A3F

4

(e) Problem with D E F

M1

∴ overrun = 1 (hour)

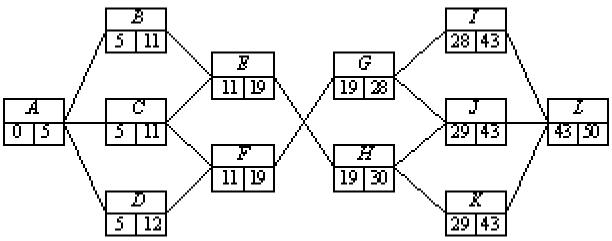
A1

2

[13]

Q10.

(a) (i)



M1

start times

A1

2

(ii) *finish times*

M1

A1

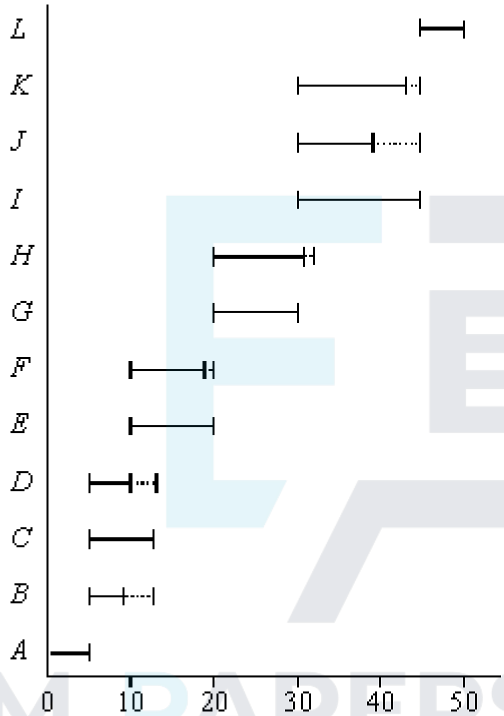
2

(b) Critical path *A C E G I L*

B1

1

(c)



M1

– 1 e.e.

A2F

3

(d) From network
Problem with *B, C, D* and *I, J, K*

M1

Minimum extra *B, J*

A1

= 50 + 1 + 6 = 57

B1

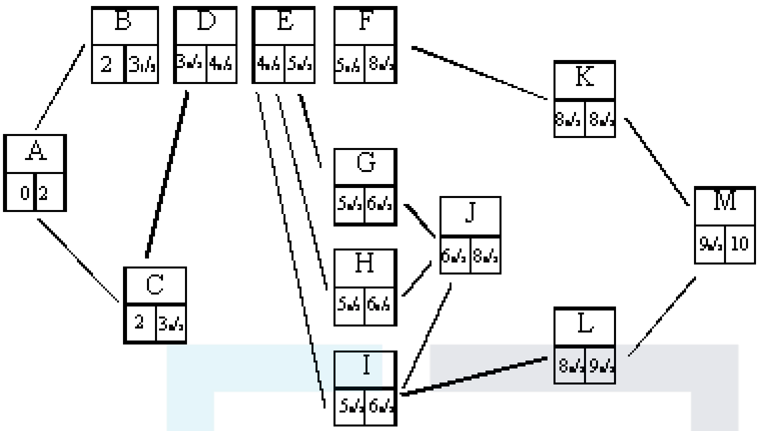
3

$$\begin{pmatrix} B + D - C = 1 \\ J + K - I = 6 \end{pmatrix}$$

[11]

Q11.

(a)



SCA

M1

– 1 each independent error

A3

4

(b) As diagram

M1 3

–1 EE

A2

(c) As diagram

M1

3

– 1 EE

A2

(d) Critical A, C, D, E, H, I, J, K, M

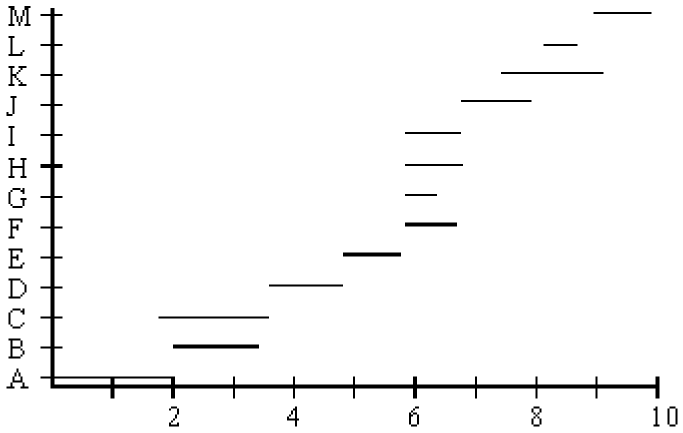
B1

10 days

B1

2

(e)



SCA

M1

–1 EE

A2f

3

(f) Electrician B, G – $\frac{3}{4}$ day

M1

Plumber C, (I) – $\frac{3}{8}$ day

A1

Joiner (H), J – $\frac{2}{5}$ day

M1

Use joiner – electric not critical

A1

New completion time $9\frac{3}{5}$

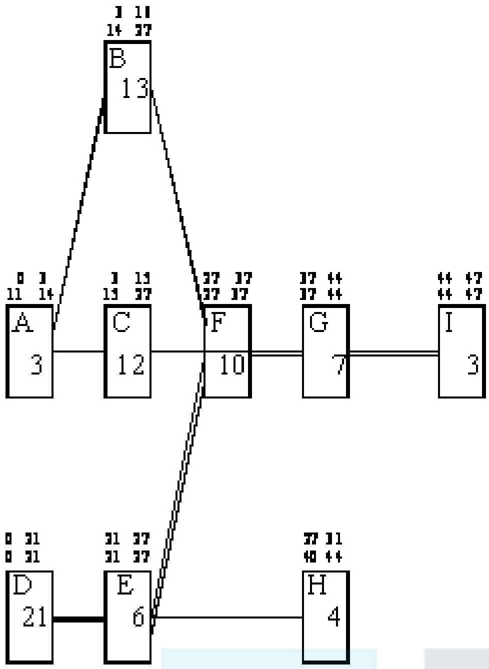
cao

5

[20]

Q12.

(a)



Attempt at a network with all 9 activities included, starting at A and D, finishing at I

M1

All correct, – 1 EE

A3

4

(b) Upper figures show earliest times

Evidenced by labels

M1

(forward scan)

All correct

A1

Lower figures show latest times

Evidenced by labels

M1

(backward scan)

All correct

A1

Critical path *D-E-F-G-I*

Critical path stated and shown. ft c's labels provided upper and lower figures match

A1FA1F

6

(c)

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*
11	11	12	0	0	0	0	13	0

– 1 EE

B2

2

(d) Cascade diagram

Attempt a cascade diagram with all activities and a suitable time scale

M1

All correct. – 1 EE

A3

Minimum number of helpers: 3

No. of helpers (can be earned independently).

B1

5

(e) Least delay: 1 day

Delay

B1

Achieved by starting *A* and *B* as early as possible
then do *C* immediately after *B*

OE involving B and C sequential rather than parallel

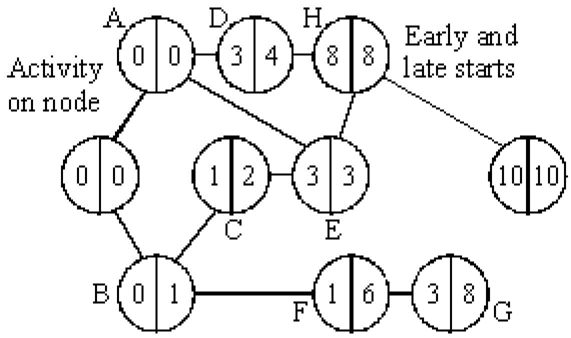
E1

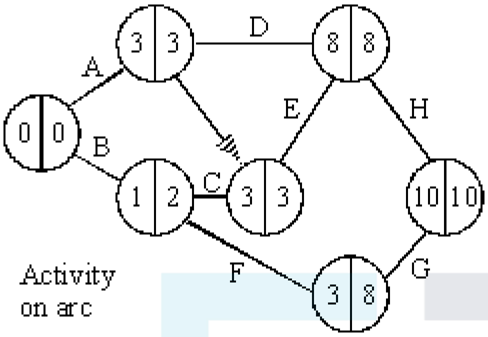
2

[19]

Q13.

(a) + (b)





time:- 10

critical activities:- A, E, H

M1

Sca

A1

single start and finish

A3

(- 1 each error)

M1

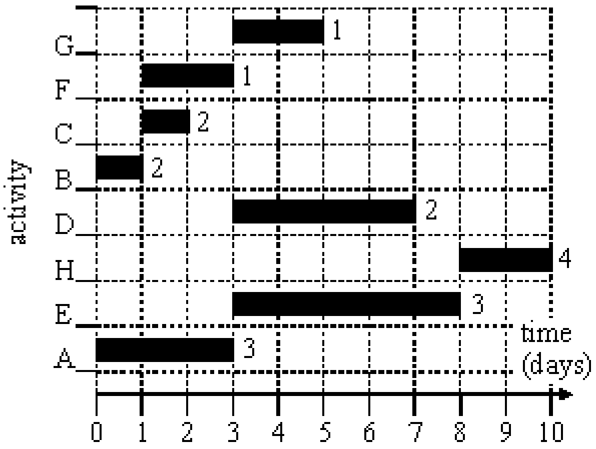
forward pass

A1 M1

backward pass

A1 B1 A1

(c)



(–1 each error)

M1 A2

(d) F – 2; G – 8; D – 4

B2 (–1)

[16]

Examiner reports

Q1.

(a) Almost every student calculated the earliest start times and latest finish times correctly.

(b) At least one critical path was found correctly by almost everyone and most students found the two critical paths.

(c) Greater care should be taken when drawing diagrams of this type, preferably ***using a ruler***. The most common error in the cascade diagram was the omission of slack for the non-critical activities; others tried to show activities starting as late as possible. Some tried to insert activity J as two sections, so occasionally the diagram resembled a histogram.

(d) Those students who made errors in their earliest start times and latest finish times for J were given credit for writing down an inequality based on their values. Quite a large number of students did not realise the need to use the difference between the latest finish time and earliest start time for J only and so had a totally incorrect inequality.

Q2.

(a) Almost every candidate calculated the earliest start times and latest finish times correctly.

(b) At least one critical path was found correctly by almost everyone and most candidates found the two critical paths. A surprising number neglected to state the minimum time for completion; it was not enough to simply have this value in the final box on the diagram.

(c) Those candidates who made errors in their earliest start times and latest finish times were given credit for identifying their activity with the greatest float time and the majority seemed aware of how to calculate its value.

(d) Greater care should be taken when drawing diagrams of this type, preferably using a ruler. The most common error in the cascade diagram was the omission of slack for the non-critical activities. Despite the bold printing of the word **late**, many showed B, E, G and J starting as early as possible. Some tried to fit activity G in two sections, with one of the two blocks slotted between activities I and J, so the diagram resembled a histogram.

Q3.

This proved to be a very high scoring opening question for most candidates.

Part (a) The earliest start times were usually correct, but the latest finish times for some activities were sometimes calculated incorrectly.

Part (b) Most candidates found at least one correct critical path, but sometimes spoiled their answers by adding a third path that was not a critical path. Some candidates listed all the critical activities instead of giving the critical paths and scored no marks for this.

Part (c) Most cascade diagrams indicated the critical activities correctly. Quite a lot of candidates failed to show the correct slack for activities C, D, G , I or J. Others had slack overlapping another activity and this was not acceptable.

Part (d) The last part of the question was answered well by most candidates, although some failed to mention the new earliest starting **day** for either K or L.

Q4.

This proved to be a very good opening question for all candidates. The earliest start times and latest finish times were usually correct. Some candidates only gave one critical path instead of two. Quite a large number thought that $ADGJK$ was also a critical path and lost a mark. Despite showing a value of 22 on their insert, several candidates forgot to state the minimum time for completion. The cascade diagram caused few problems. Some candidates used blocks with critical activities linked together and other activities drawn in blocks above; others used a sequence of horizontal lines with the vertical axis labelled from A to K. The main reason for losing marks in part (c) was the ignoring of the slack or float. The explanation in part (c) was not always adequate: it was necessary to explain that F needed to start after 12 days at the earliest, with I and K being delayed by one day, and hence that the new minimum completion time was 23 days.

Q5.

The network diagram and the critical path were answered very well by all candidates. A few were unable to identify float times for the non-critical activities. The major problem in the Gantt chart was a failure to show slack time. When this was indicated it was usually shown after the various activities even though the question asked to show activities starting as **late** as possible. Some indication should have been made on the diagram to indicate that G needed to start after F and also after C but this was not penalised this time.

Q6.

This question proved to be a good source of marks for all candidates, although a significant number failed to score full marks in part (d) due to the inaccuracies of their diagram or through drawing a resource histogram. Part (e) discriminated well between candidates. This was the first time that candidates were provided with a stage/state table for answering a dynamic programming question. The question gave candidates the opportunity to use a network diagram and still score full marks. A significant number of weaker candidates thought that, as there were only six possible orders, a complete enumeration was an acceptable method. This was not the case. About half of the candidates worked backwards through the network and about half worked forwards. Both methods were equally successful.

Q7.

This question proved to be a good source of marks for all candidates, although a significant number of candidates failed to score full marks in part (c), due to the inaccuracies of their diagram or through drawing a resource histogram instead of a Gantt diagram.

Q8.

This question was well answered by candidates. Marks were lost in the final part of the question where candidates did not show the float times, where appropriate. Nearly all candidates used their answer book for their Gantt diagram and full marks were regularly obtained, provided the correct timings were clearly illustrated.

Q9.

This question was well answered, with the majority of candidates scoring at least 10 out of 13 marks. Whereas parts (a), (b) and (c) were nearly always correct, candidates did make mistakes in part (d). The question stated that activities were to start as **late** as possible. This was the first time that this had been asked. A number of candidates made a mistake with activity E.

Also, for the first time, some candidates failed to show accurately the start and finish times for their Gantt diagram. If candidates draw their answer in their answer booklet, they must clearly indicate start and finish times for each activity. Candidates who used graph paper did not lose accuracy marks.

Q10.

Parts (a), (b) and (c) were answered well by many candidates but part (d) proved to be the hardest question on the paper. There were some mistakes made on the backward pass to find the latest finish time for the activities. The majority of candidates used graph paper to draw their Gantt diagram, a practice that should be encouraged. In part (d) those candidates who looked at the original network realised that with only two workers there was a problem with B, C, D and I, J, K and found the minimum extra time that two workers would need to complete the three activities and arrived at the correct solution.

Q11.

The question involved a routine application of critical path analysis. The candidates were well prepared and scored nearly all the available marks in the first five parts. It was pleasing to note that candidates worked their critical path on their network diagram, with the benefit of saving time for the later, more searching, parts of the question. Although part (f) was challenging, there were a number of correct solutions with candidates able to relate the new information back to the critical path together with its implications.

Q12.

The question was a routine application of critical path analysis for which the candidate was well prepared and scored nearly all the available marks.

Q13.

This question was generally very well answered, with many candidates obtaining at least 14 of the available 16 marks. Almost everyone obtained the correct activity network, and almost all of the forward passes were correct. There were some slips made in the backward pass, particularly for activity B. Most obtained the correct minimum duration and list of critical activities.

In part (c) the majority of candidates completed the cascade chart correctly, although a few showed all of the activities starting at time zero. Far fewer could obtain the new scheduled start times in part (d).